



Amoxicillin functionalized gold nanoparticles reverts MRSA resistance

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ABSTRACT

In this study, we have described the biosynthesis of biocompatible gold nanoparticles (GNPs) from aqueous extract of the aerial parts of a pteridophyte, "*Adiantum philippense*" by microwave irradiation and its surface functionalization with broad spectrum beta lactam antibiotic, amoxicillin (Amox). The functionalization of amoxicillin on GNPs (GNP-Amox) was carried out via electrostatic interaction of protonated amino group and thioether moiety mediated attractive forces. The synthesized GNPs and GNP-Amox were physicochemically characterized. UV–Vis spectroscopy, Zeta potential, XRD, FTIR and SERS (surface enhanced raman spectra) results confirmed the loading of Amox into GNPs. Loading of Amox to GNPs reduce amoxicillin cytotoxicity, whereas GNPs were found to be nontoxic to mouse fibroblast cell line (L929) as evident from MTT and acridine orange/ethidium bromide (AO/EtBr) live/dead cell assays. The GNP-Amox conjugates demonstrated enhanced broad-spectrum bactericidal activity against both Gram-positive and Gram-negative bacteria. Furthermore, *in-vitro* and *in-vivo* assays of GNP-Amox revealed potent anti-MRSA activity and improved the survival rate. This indicates the subversion of antibiotic resistance mechanism by overcoming the effect of high levels of β -lactamase produced by methicillin resistant *Staphylococcus aureus* (MRSA). Taken together, this study demonstrates the positive attributes from GNP-Amox conjugates as a promising antibacterial therapeutic agent against MRSA as well as other pathogens.

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1. Introduction

The worldwide escalation of mortality rates due to antibiotic resistant bacterial strains poses a serious challenge to modern medicine and therapeutics. According to World Health Organization (WHO), it is estimated that patients infected with methicillin resistant *Staphylococcus aureus* (MRSA) are 64% more likely to die compared to people infected with non-resistant form of the bacteria [1]. Therefore, there is an urgent need for new improvements in current methods and novel strategies are required to cope with the emergence of antibiotic resistant bacterial strains. Recent studies have demonstrated the beneficial properties of combination of nanoparticles with antibiotics for enhancement of their bactericidal properties and also reduction of the cytotoxicity of both components. Currently, nanomaterial based platforms for delivery of antibiotics has escalated interest as it restores the antibiotic-efficacy against drug resistant bacteria by providing increased concentration of antibiotics at the site of bacterium–antibiotic interaction and facilitating its binding to bacteria [2]. Though, gold nanoparticles (GNPs) are generally regarded as biologically inert, but conjugation with antibiotics, the antibacterial effectiveness of

GNP-antibiotic conjugates increases by many folds. Similar trend has been reported by some research groups using GNPs, conjugated with vancomycin as well as aminoglycoside antibiotics [3,4,5]. There are reports where, internalized amino-substituted pyrimidines capped GNPs have been postulated to induce disruption of bacterial cell-wall by sequestration of Ca^{2+} and Mg^{2+} ions and eventually retardate the development of drug resistance by inhibiting DNA and protein synthesis [6]. However, enhanced anti-bactericidal activity can only be possible with formation of a stable GNP-drug conjugate and availability of sufficient concentration of antibiotics on the bacteria-antibiotic site of interaction [7]. Development of new strategies for proper functionalization of the nanoparticles for strong and efficient drug loading is one of the most critical steps for GNP mediated drug delivery. Currently, different methods available for functionalization of GNPs as a drug delivery vehicle include systems based on covalent binding, drug encapsulation, electrostatic adsorption and other non-covalent assemblies [8–16].

In the present study, we have synthesized biocompatible GNPs using plant extracts through microwave (MW) irradiation. MW is a fast emerging and widely accepted processing technology for a variety of synthesis and biomedical applications [17,18,19,20,21,22,23]. Compared to the conventional heating, MW irradiation shortens reaction times and improves the yield of nanoparticles production without causing any appreciable alteration in its composition. Plant extracts,

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mainly angiosperms, have been widely used for reduction and capping of nanoparticles, while Pteridophytes largely remain unexplored in this context [24]. Pteridophytes (ferns and fern allies) are one of the oldest known terrestrial plant groups on earth and we have explored one of the most highly valued fern, "*Adiantum philippense* L. (*Adiantum lunulatum* Burm. F.) (family: Adiantaceae) for synthesis of gold nanoparticles (GNPs). The plant has been utilized by different communities as demulcent, emollient, expectorant, emmenagogue etc. The plant also controls bacterial infections of spleen and liver. Furthermore, it is also useful in treatment of inposthumes (purulent swelling and other abscess) of uterus, and against tuberculosis and influenza [25]. There are also a few reports of antimicrobial activity of organic and aqueous extracts of *A. philippense* [26].

The synthesized gold nanoparticles were further functionalized with broad spectrum β -lactam antibiotic, amoxicillin by electrostatic interaction between negatively charged nanoparticles and protonated amino groups of the antibiotic. As the thioether moiety of amoxicillin is prone to interact with transition metals like gold that also influences adsorption of this drug on the GNPs surface.

The present study is aimed at investigating whether attachment of amoxicillin to GNPs could enhance the efficacy of the antibiotic against selected pathogenic Gram-positive and Gram-negative bacteria, namely *S. aureus*, *Staphylococcus epidermis*, *Escherichia coli*, *Bacillus subtilis*, *Bacillus cereus* and four MRSA clinical isolates. Amoxicillin (Amox), a broad-spectrum antibiotic has been used as a model drug in this study, which acts against a wide range of disease-causing bacteria and also used extensively in pediatric health care. This study focuses mainly on understanding the effect of GNP-Amox conjugate on restoring the amoxicillin activity against MRSA clinical isolates.

2. Experimental

2.1. Materials

Healthy aerial parts of the fern *A. philippense* were collected from a place near to IIT Guwahati. The fern was collected from a massive colony growing on a roadside wall (Fig. S1). Taxonomic identification of the sample was done at Department of Botany, Gauhati University, Assam, India and a voucher specimen (no. 16953) was deposited in the herbarium library. Bacterial strains were cultured on nutrient agar plates and maintained in the nutrient agar slants at 4 °C. The organisms tested were *S. aureus* (MTCC 3160), *S. epidermis* (MTCC 425), *E. coli* (MTCC 40), *B. subtilis* (MTCC 441), *B. cereus* (ATCC 10702) were either obtained from IMTECH, Chandigarh, India and ATCC, Manassas, USA. Four numbers of methicillin resistant *S. aureus* strains (MRSA 01-04) were collected from Hayat Hospital, Guwahati, Assam, India. Clinical isolates were obtained from patients with written consent according to standards of the institutional review board at Institute of Advanced Study in Science and Technology (IASST, Guwahati). Identification of MRSA was confirmed according to the recommendations of the National Committee for Clinical Laboratory Standards. For cytotoxicity studies, mouse fibroblast cell line (L929) was procured from NCCS, Pune, India.

2.2. Synthesis and characterization of GNPs

Microwave mediated green synthesis of GNPs were performed by utilizing aqueous extract of *A. philippense* (AE) and physicochemical characterizations were done as per established methods in our laboratory [27,28]. Briefly, leaves were washed twice with tap water and double distilled water (dd.H₂O) respectively, shade dried in dust free condition, grinded into fine powder followed by extraction in dd.H₂O at a concentration of 1 g per 50 ml after 72 h of shaking incubation at 25 °C. The final filtered aqueous extract (AE) was employed for the synthesis of GNPs. Different volumetric concentration (1–10%) of AE was mixed with 1 mM fixed concentration of HAuCl₄ aqueous solution and the resultant mixture solution was subjected to microwave irradiation in a

domestic microwave oven (900 W, 2.45 GHz, LG MO- MC-767 W/WS). 5 ml of final reaction volume was maintained in all experiments. Reactions were stopped at the change of original yellowish colour into different hues of red colour. Time was noted for each applied concentration of AE. Qualitative tests for different phytochemicals (terpenoids, alkaloids and flavonoids) of the plant extract and the synthesized GNPs were also performed according to well established protocols [29].

Ultraviolet-visible (UV-Vis) spectroscopic measurements were carried out on a Carry 100 BIO UV-Vis spectrophotometer (Varian, CA, USA). Morphology and size and purity of the synthesized GNPs were investigated using JEOL 2100 Ultra High Resolution Transmission Electron Microscopy (UHR-TEM) and Field Emission Scanning Electron Microscopy—Energy—dispersive X-ray spectroscopy (FESEM-EDX) analysis. Zeta potential and particle size analysis were performed with dynamic light scattering (DLS) measurements (Malvern Instruments Ltd., Malvern, UK). X-ray powder diffraction (XRD) was carried out with a Bruker D8 ADVANCE X-ray powder diffractometer (Bruker AXS Inc.). Fourier transform infrared spectroscopy (FT-IR) of purified and lyophilized AE and GNPs were done on an infra-red spectrophotometer (Thermo Scientific Nicolet Is 10).

2.3. Amoxicillin loading on gold nanoparticles

25 mg/ml of amoxicillin with appropriate amount of 2-(N-morpholino) ethanesulfonic acid (MES) buffer at pH 6 was added in GNPs solution (25 mg GNPs dispersed in 10 ml Milli-Q water) [30]. The solution was then incubated at room temperature under mild magnetic stirring and changes in surface plasmon resonance (SPR) were monitored by UV/Vis spectroscopy measurements. IR spectra of amoxicillin bound to the GNPs were recorded and surface charge of GNPs before and after loading of amoxicillin was determined by measurement of zeta potential. XRD measurements were done to confirm the drug nanoparticle interaction. SERS spectra of bare GNP and GNP-Amox were recorded using Laser micro raman system and excited using the 785 nm excitation laser at 25.7 mW (Horiba Jobin Vyon, Model Lab Ram HR). The incident laser beam was focused and the signal collected using a 20× long distance objective.

Unbound Amox molecules were removed by centrifugation at 30,000 g for 10 min. To evaluate the amoxicillin-loading efficiency, the content of the residual Amox in the supernatant was determined by UV-Vis measurements [31]. The Amox-loading efficiency was calculated by

$$\text{Loading efficiency(\%)} = \frac{\text{Total Amox} - \text{Amox in supernatant}}{\text{Amox in supernatant}}.$$

2.4. Cytotoxicity studies

For cytotoxicity study of GNPs, MTT (3-[4, 5-dimethylthiazole-2-yl]-2, 5-diphenyl tetrazolium bromide) dye conversion assay, Acridine orange–ethidium bromide (AO–EtBr) live/dead cell assay and colony forming assay were carried out on mouse L929 fibroblastic cell line (obtained from NCCS, Pune) [32,33].

2.4.1. MTT assay

Cells were cultured and maintained according to supplier guidelines. L929 cells at a density of 1×10^4 per well were cultured in a 100 μ l volume of cell culture medium (DMEM: Dulbecco's Modified Eagle Medium) supplemented with 10% fetal bovine serum in a 96-well cell culture plate. After 24 h, cultured cells were treated with a series of different concentrations (5, 10, 20, 40, 50, 100, 200, and 250 μ g/ml) of filter sterilized GNPs in 100 μ l/well EMEM without serum and incubated further for 24 h. This was followed by removal of the media and treatment with MTT dye at a final concentration of (0.5 mg/ml) and further incubated for 4 h. Finally, 100 μ l of dimethylsulfoxide (DMSO) was added

to each well to dissolve blue formazan precipitate, and absorbance was measured at 570 nm using a microplate reader (Bio-Rad Model 680; Bio-Rad). The cell viability was expressed as a percentage of the control by the following equation.

$$\text{Viability(\%)} = \frac{N_t}{N_c} \times 100$$

where N_t is the absorbance of the cells treated with GNPs and N_c was the absorbance of the untreated cells ($n = 5$; where n is the no. of independent experiments).

2.4.2. AO–EtBr assay

AO–EtBr live/dead cells staining of untreated L929 cells and those cells treated with the lowest (5 µg/ml) and the highest (250 µg/ml) concentrations were done for qualitative analysis of biocompatibility of GNP-Amox. Treated cells were observed under an inverted microscope (Eclipse TS100, Nikon) equipped with fluorescence unit and digital camera (Coolpix 5400, Nikon).

2.4.3. Colony forming assay

Colony forming assay is one of the established method to perform cell viability assessment. In this assay, the L929 cells were treated with different concentrations of the bare GNP, GNP-Amox and free Amox (5 µg/ml to 250 µg/ml) and were trypsinised. Then 1000 cells were counted (Automated cell counter, Invitrogen, USA) and seeded in 25 cm² cell culture flasks. Untreated L929 cells were treated with PBS and kept as negative control. Then the cells were kept for incubation period of 10–12 days and evaluated the score for viable colonies. Then 0.25% methylene blue in 75% carbinol was used to fix the colonies. Thereafter only those colonies were counted which have at least 50 or more cells. The plating efficiency (PE) was calculated as the percent number of cells which grow into colonies in control group for normalizing the colony counts from treated group. Surviving fraction (SF) is calculated as colonies counted divided by the number of colonies plated with a correction for the plating efficiency. The experiment was performed in triplicate for all the three samples.

2.5. Zone of inhibition (ZOI), minimum inhibitory concentration (MIC), minimum bactericidal concentration (MBC)

The *in-vitro* antibacterial screening was done by agar well diffusion method [34]. Briefly, 200 µl of tested bacterial suspensions were uniformly spread over solidified nutrient agar plates with the help of a sterilized spreader. Wells of 6 mm diameter were made in the centre of these agar plates with the help of a sterile cork borer. The wells were then filled with 25 µg/ml of GNP, free Amox or GNP-Amox and allowed to diffuse at room temperature for an hour. Then the plates were incubated at 37° ± 2° for 24 h. The antibacterial activities of the test solutions were determined by measuring the diameter of the inhibition zone around the well.

The increase in antibacterial activity of GNP-Amox compared to free Amox against the tested bacteria were quantified by the equation $(B^2 - A^2) / A^2 \times 100$, where A and B are the ZOI for free Amox and GNP-Amox, respectively.

Minimum inhibitory concentration (MIC) was evaluated for GNP-Amox and free Amox against nine bacterial strains by the visual broth macro dilution method using nutrient broth medium [35]. As bare GNPs did not show any zone of inhibitions, so we did not perform MIC/MBC experiment for GNPs. Minimum bactericidal concentration (MBC) was determined using standardised method [36]. The tube of GNP-Amox and free Amox, which showed no visible growth after 48 h of incubation when sub cultured on to samples free nutrient agar plated using an inoculum size of 0.5 ml was considered to be the MBC.

2.6. In-vitro anti-biofilm assay

The effect of GNP-Amox on biofilm formation was assessed by a modified microtiter plate assay [37]. Sterile polystyrene microtiter plate wells were inoculated with 200 µl of nutrient broth containing 10⁶ [6] colony forming unit (CFU) ml⁻¹ of tested bacterial culture and loaded with 16 µg ml⁻¹ of GNP-Amox. The plates were incubated in static mode at 37 °C for 48 h. The negative and positive control wells were maintained without GNP-Amox and amoxicillin-treated bacterial suspensions, respectively. After 48 h of growth, the medium in the wells was removed and washed with sterile PBS (pH-7.4) to remove loosely attached bacteria. Further, staining of wells were done with 200 µl of 0.1% crystal violet and incubated for 30 min followed by washing and air drying. Bound stain was solubilized in 200 µl of 95 % ethanol, and absorbance was measured at 590 nm.

2.7. Amoxicillin and GNP-Amox kill test

To determine the rate at which the free Amox and GNP-Amox killed MRSA, kill test was performed [38]. Resistant strains were selected based on MIC/MBC tests. Strains were grown overnight at 37 °C with shaking in nutrient broth. The next day, the bacteria were washed twice in PBS (centrifugation speed 13000 × g), added to nutrient broth to achieve a final concentration of approximately 2 × 10⁵ CFU/ml and then incubated at 37 °C with shaking. Free Amox and GNP-Amox were added to bacterial cultures at a concentration of 16 µg/ml (selected based on MIC results). Samples were taken at 0, 2, 4 and 6 h and analyzed to determine the number of viable bacteria, which remained after treatment. The CFU of MRSA recovered post treatment were compared for both free Amox and GNP-Amox and plotted as the CFU/ml over time post treatment. Three independent experiments were run in triplicate. Comparisons were made using an unpaired Student's t-test.

2.8. In vivo treatment of systemic MRSA infection

An overnight MRSA (MRSA 01) culture was diluted in Nutrient broth, grown to an OD₆₀₀ of 0.6 at 37 °C with shaking, centrifuged and then washed twice in PBS. Albino male mice (eight mice per group; age: 4 weeks; weight: 20 g) were infected intraperitoneally with 3.9 × 10⁶ CFU of the tested MRSA. One hour after infection the mice were injected intraperitoneally with PBS, 50 mg/kg of GNP-Amox and 50 mg/kg of free Amox. After 24 hours post treatment, four mice from each group were sacrificed and the spleens and kidneys were harvested, homogenized and plated on agar for enumeration of CFU. The remaining four mice of each treatment group were given the above mentioned dose by duration gap of every 24 hours and survival rate was determined and expressed by Kaplan–Meier curves (daily over a 7-day period). The animal infection experiment was approved by the Institutional Animal Ethical Committee of IASST on the use and care of animals and performed using accepted veterinary standards.

2.9. Determination of gold concentration in kidney and spleen

Kidney and spleen tissue samples harvested after 24 h post treatment were digested following previously reported procedures [39]. Briefly, 5 g of each homogenized kidney and spleen sample tissues were dried completely in a clean oven at 60–70 °C. Each dried tissue was digested to white ash using 30% hydrogen peroxide at 50–60 °C followed by digestion with concentrated 0.25 N ultra-pure trace-metal free nitric acid (1:2–1:50 v/v). Estimation of gold concentration in the digested tissue samples were performed by using Graphite furnace atomic absorption spectrophotometry (GFAAS) (Perkin–Elmer Model-pinnacle 900 H). Elemental gold standard (Sigma Aldrich Chemical Co.) was used for all calibration curves. All GF-AAS analyses were performed at a temperature of 1800 °C.

3. Results

3.1. Synthesis and characterization of GNPs

In this study, GNPs were synthesized via microwave irradiation. There was change in colour of the yellow coloured mixed aqueous solutions containing AE (1–10%, v/v) and HAuCl_4 (1 mM) into different ranges of red hues due to microwave irradiation, which indicates the formation of polydispersed GNPs (Table S1). Herein, auric ions (Au^{3+}) were reduced by the AE into the gold atoms (Au^0) followed by formation of GNPs by aggregation). The phytochemicals like terpenoids, flavonoids and alkaloids class of molecules act both as reducing and stabilizing agent for the synthesis of stable GNPs.

The UV–Vis analysis of the MW irradiated coloured solutions exhibited very distinct surface plasmon resonance (SPR) absorption spectra each with a sharp SPR peak centered at different positions showing both blue and red shift tendencies (Fig. S2). The maximum blue shifted SPR peak was centered at 517 nm for the SPR spectrum obtained at 7% (v/v) of AE. For all types of qualitative instrumental analysis and cytotoxicity studies, we took GNPs synthesized at 7% (v/v) of AE, as it resulted in highest product yield with narrow SPR peak among all the applied concentrations of AE. Furthermore, the morphology and size of the GNPs were characterized using TEM analysis, which revealed a mixed population of spherical and triangular shaped GNPs of various sizes (Fig. 1A). Selected-area electron diffraction

pattern (SAED) and ultra-high resolution TEM (UHRTEM) analysis of a single spherical particle confirmed its crystalline nature (Fig. 1B, C). A histogram representing the size distribution (Fig. 1D) of GNPs corresponding to TEM image revealed the average particle size to be around 33.9 ± 14.0 nm. FESEM-EDX result confirmed the synthesis of GNPs (Fig. S3). XRD patterns of GNPs exhibited four prominent Bragg reflections with 2θ values at around 38.4° , 44.5° , 64.7° and 78.01° of lattice planes and were indexed to the (1 1 1), (2 0 0), (2 2 0) and (3 1 1) facets of gold respectively (Fig. S3). Characteristic FTIR bands of AE showed presence of hydroxyl ($-\text{OH}$), alkanes, aliphatic amines and aromatic amines, $\text{C}=\text{C}$ (benzene origin) and carbonyl functional groups (Fig. S4, curve A). Almost all IR bands present in AE were also observed in the GNPs sample with a considerable shift in transmittance level (Fig. S4, curve B). The capping of GNPs by the phytochemicals present in AE is important as it stabilizes the nanoparticle and also the phytoconstituents are mainly responsible for the medicinal properties possessed by a plant. The qualitative tests done for phytochemicals of the plant extract showed the presence of terpenoids, flavonoids and alkaloids in AE. More importantly the synthesized GNPs also exhibited the presence of the same phytochemicals as found in the plant extract. This suggest that phytochemicals like terpenoids, flavonoids and alkaloids class of molecules act both as reducing and stabilizing agent for the synthesis of stable GNPs. Therefore, capped GNPs carrying these plant molecules on their surfaces may be useful for different biomedical applications.

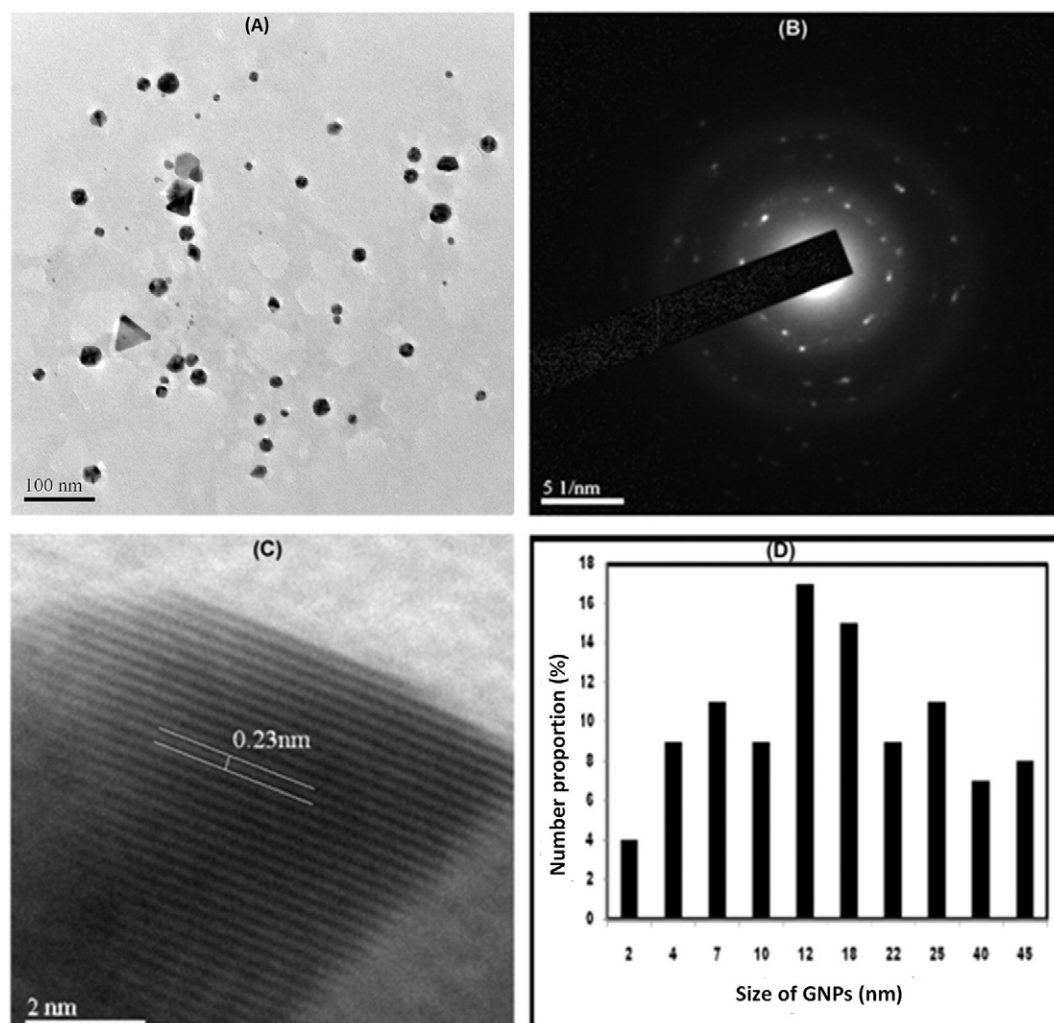


Fig. 1. TEM (A), SAED patterns (B), HRTEM (C) and size distribution (D) of the synthesized GNPs.

3.2. Amoxicillin loading on gold nanoparticles

During the biosynthesis of GNPs, different plant phytochemicals self assembled on the surface of GNPs which imparts negative charge to nanoparticles. Amoxicillin has a protonated amino group through which it can easily adsorb to metallic nanoparticles carrying net negative charge on its surface [15,16]. Again the thio-ether moiety present in amoxicillin has a strong attraction towards the transition metal like the GNPs. The wavelength of GNPs shifted considerably from 517 nm (7%) to 528 nm after conjugated with amoxicillin (data not shown). The change of the absorption spectrum might be due to the presence of protonated nitrogen atom of amoxicillin molecule and combination of positively charged amine group on the surface of GNPs. The near-field excitation shows large increase in SERS spectral intensity with a small difference in wave number of GNP-Amox due to amoxicillin adsorption to the GNPs (Fig. S5). The zeta potential of GNPs was -28 mV compared to zeta potential of GNP-Amox i.e. -22.1 mV (Fig. S6). This decrease in the zeta potential may be attributed to the presence of positively charged amoxicillin and it also indicates that attractive forces including hydrogen bonds could be playing a vital role in facilitating the drug-loading process [40]. Loading efficiency of Amox to GNP was calculated 77 ± 4 percentage. So one mg of gnp contains 770 μ g of amoxicillin. It is well known that the surface properties of colloidal systems are critical in determining their drug carrier potential, since they control their interactions with plasma proteins. Zeta potential measurements provide information about the overall surface charge of the particles and how this is affected by changes in the environment (e.g. pH, presence of counter-ions and adsorption of proteins). XRD analysis showed enhanced intensity of Bragg reflections of GNP-Amox compared to free amoxicillin due to the increased crystalline nature of GNP-Amox (Fig. 2). These results demonstrate the interaction between GNP and amoxicillin. The bonding between protonated amine groups of the amoxicillin molecule with surface of GNPs was also confirmed by FTIR (spectra not shown). Amoxicillin showed N–H stretching peak of amine at 3777 cm^{-1} . In case of GNP-Amox conjugate, the amine peak shifted to 3373 cm^{-1} which signifies the formation of electrostatic bond formation. Since electrostatic interactions are strongly affected by solution pH, any changes in pH can cause nanoparticles aggregation. Thus it is very much required to select an appropriate buffer system. In our study, we found 2-(N-morpholino) ethanesulfonic acid (MES) buffer (pH 6) is suitable, which protect aggregation of GNPs after addition of amoxicillin to the reaction mixture.

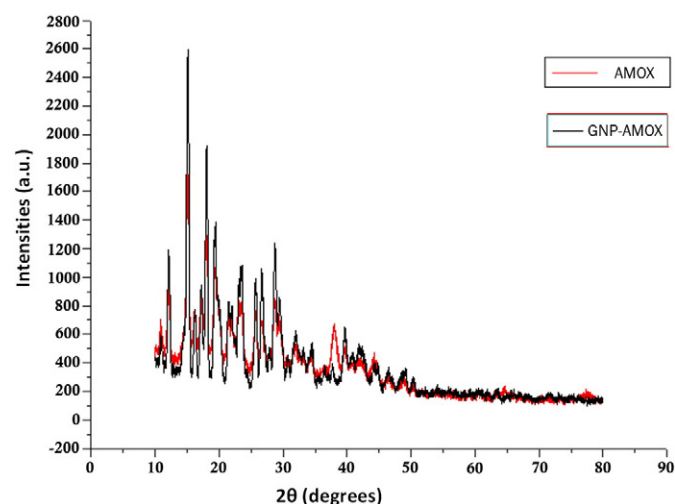


Fig. 2. Comparative XRD pattern of free Amox and GNP-Amox.

3.3. Cytotoxicity analysis

In the MTT cytotoxicity assay, after 24 h of post treatment, L929 cells showed high viability (81%) even up to 250 μ g/ml concentration of GNPs (Fig. S7). This was a good response of the used cell line against the GNPs. Previously it has been reported that a vast majority of gold (I) and gold (III) compounds exhibit varying degrees of cytotoxicity to a variety of cells [41,42]. It was also observed that GNP-Amox exhibited a much higher cell viability (79%) compared to free-Amox (77%). Prior to any *in vivo* application of the *A. philippense* mediated GNPs and GNP-Amox, the high dose tolerance exhibited by a mammalian cell line is very important. The present green route of GNPs synthesis employing AE of the fern *A. philippense* and MW irradiation thus resulted in biocompatible GNPs as well as GNP-Amox against L929 cell line.

AO–EtBr is fluorogenic reagents that stain cells based on the uptake or exclusion of these dyes by the treated cells. The basic principle of AO–EtBr cell staining lies in intake (live cells) or loss (dead cells) of cell membrane integrity. AO is membrane-permeable that stains both live and dead cells, while EB is only able to pass through the membrane of a dead or dying cell. Each dye that is taken up by a cell fluoresces—AO makes a cell green, and EB makes a cell red [43]. L929 cells treated with the lowest (5 μ g/ml) and the highest (250 μ g/ml) concentrations of GNP-Amox retained their viabilities. The L929 cells treated with the highest (250 μ g/ml) concentrations of GNP-Amox emitted bright green fluorescence (Fig. 6B) and very weak red fluorescence (Fig. 6C) due to AO inclusion and EB exclusion phenomenon respectively. The normal microscopic view of the same treated cells revealed the intake of morphology (Fig. 6A).

Cytotoxicity analysis by colony forming assay revealed that treatment with bare GNP, GNP-Amox and free Amox on L929 cell line for 6–48 h up to 200 μ g/ml does not induce any toxicity as evident from the survival fraction values of 94.2, 96.2 and 92.6 respectively for the tested samples.

3.4. In-vitro antibacterial activity of GNP-Amox and free Amox

GNP-Amox was found to enhance antibacterial activity of amoxicillin against the tested pathogens. More interestingly, upon loading to GNPs, amoxicillin restore its antibacterial activity against all the MRSA clinical isolates. More specifically, MRSA1 exhibited the highest increase in ZOI ($p \leq 0.004$), whereas minimal increase in fold area in case of *S. epidermis* (Table 1). It is worth mentioning that, bare GNPs did not exhibit any antibacterial activity. GNP-Amox exhibited significantly lower MIC/MBC values than that of free Amox against the tested bacterial pathogens ranging from $1/2$ (μ g/ml)– $16/32$ (μ g/ml), whereas GNP-Amox rescue the antibacterial activity of free Amox demonstrated MIC/MBC values ranging from $16/32$ (μ g/ml)– $32/64$ (μ g/ml) (Table 2). The free Amox didn't show MRSA growth inhibition up to 5 mg/ml of concentration. The GNP-Amox was able to impede the biofilm

Table 1

Mean zone of inhibition of Amoxicillin and GNP-Amox against various human pathogenic bacteria including MRSA strains.

S.no	Bacterial strains	Amoxicillin (zone of inhibition) (mm) ^a	GNP-Amox (zone of inhibition) (mm) ^a	Increase in fold area %
1	<i>E. coli</i>	19 ± 0.64	31 ± 0.99	62
2	<i>Staphylococcus aureus</i>	25 ± 0.41	30 ± 0.75	31
3	<i>Staphylococcus epidermis</i>	21 ± 0.55	19 ± 0.38	18
4	<i>Bacillus subtilis</i>	23 ± 0.61	35 ± 0.77	57
5	<i>Bacillus cereus</i>	24.5 ± 0.49	38 ± 1.06	58
6	MRSA1	5 ± 0.21	16 ± 0.41	90
7	MRSA2	7 ± 0.26	15 ± 0.31	78
8	MRSA3	7 ± 0.19	12 ± 0.28	66
9	MRSA4	8 ± 0.19	12 ± 0.26	56

^a Values are mean $\pm$ S.D. (n = 3).

Table 2

MIC/MBC of free Amoxicillin and GNP conjugated Amoxicillin (GNP-Amox) against ten human pathogenic bacteria including MRSA strains.

S.no	Test bacteria	GNP-Amox MIC/MBC (μg/ml)	Amoxicillin MIC/MBC (μg/ml)
1	<i>E. coli</i>	2/4	6/8
2	<i>Staphylococcus aureus</i>	1/2	2/4
3	<i>Staphylococcus epidermis</i>	1/2	2/4
4	<i>Bacillus subtilis</i>	16/32	64/64
5	<i>Bacillus cereus</i>	8/16	64/64
6	MRSA1	16/32	–
7	MRSA2	16/32	–
8	MRSA3	16/32	–
9	MRSA4	32/32	–

formation by 94%, 93%, 93% and 89% respectively for the tested MRSA 1, MRSA 2, MRSA 3 and MRSA 4 strains, whereas in case of free-Amox, the inhibition against the same strains were 75.3%, 87.4%, 82.4% and 77.9% respectively. Since MRSA1 and MRSA2 were found to be most susceptible to GNP-Amox, so these were taken for further study the killing kinetics. GNP-Amox demonstrated enhanced killing of MRSA 1 and MRSA 2 compared to free amoxicillin in 2, 4 and 6 h time points (Fig. 3). The difference between the GNP-Amox and free-Amox groups remained significant for the dose of 16 μg/ml against both the tested MRSA strains.

3.5. Efficacy of gold-Amox in a systemic MRSA murine infection model

As MRSA 1 was found to be most susceptible to GNP-Amox, therefore we evaluated GNP-Amox against systemic MRSA1 infection model. At 24 h post-infection, and 23 h of post treatment the CFU burden in the kidneys and spleen of the experimental animals were enumerated (Fig. 4). Overall, GNP-Amox showed reduction in the bacterial burden by 2–3 logs in the spleen and kidneys compared with the PBS group ($p \leq 0.001$). GNP-Amox treatment reduced the bacterial load in kidneys

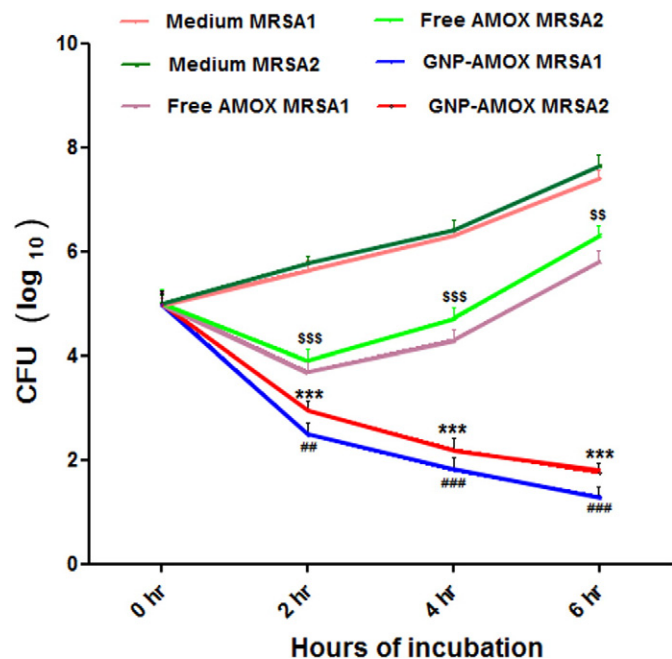


Fig. 3. Time-kill assay of (MRSA-1) ATCC 43300, (MRSA-2) ATCC 29737 was incubated with free or GNP-AMOX. Surviving cfu at selected time points are shown. All the results were expressed in mean $\pm$ S.D. ($n = 3$). \$\$\$ $p < 0.001$, \$\$\$ $p < 0.01$, comparison of Medium MRSA1 and Medium MRSA2 with free AMOX MRSA1 and free AMOX MRSA2 at different time points. *** $p < 0.001$ comparison of Medium MRSA1 and Medium MRSA2 with GNP-AMOX MRSA1 and GNP-AMOX MRSA2 at different time points. ## $p < 0.01$, ## $p < 0.01$ in comparison of free AMOX MRSA1 and free AMOX MRSA2 with GNP-AMOX MRSA1 and GNP-AMOX MRSA2 at different time points.

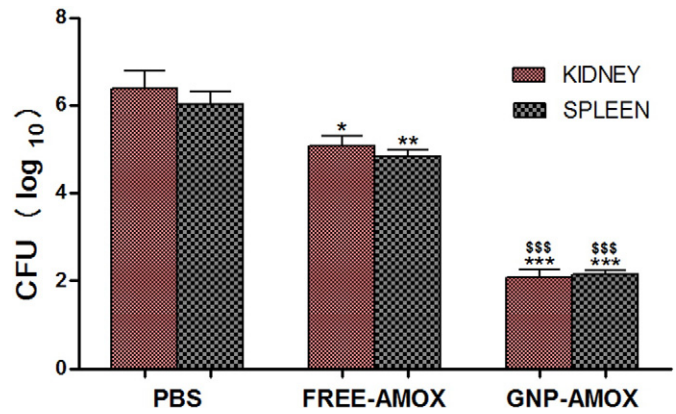


Fig. 4. MRSA bacterial burden on spleen and kidney of mice treated with PBS, free Amoxicillin (Amox) and GNP-AMOX after infected with 3.9×10^6 CFU of MRSA (ATCC 43300). All the results were expressed in mean $\pm$ S.D. ($n = 6$). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ in comparison of PBS treated animals with free-AMOX and GNP-AMOX treated animals. \$\$\$ $p < 0.001$ in comparison of free-AMOX treated animals with GNP-AMOX treated animals.

by 2–3 log and spleen by 1–2 log (approximately), which was statistically significant both for the kidney CFU ($p = 0.011$), and for the spleen CFU ($p = 0.013$) compared to free amoxicillin treatment. The survival rate at day seven post-inoculation was higher in the GNP-Amox group (96%) compared with the free Amox group (64%) and the control group (all dead within 2 days) (Fig. 5).

3.6. Distribution of GNPs and Amox in organs

There was considerable amount of gold accumulation in kidney and spleen. After 24 h of treatment, GF-AAS analysis showed 1.5 μg GNP/g tissue of kidney and 2.1 μg GNP/g tissue of spleen. Since amoxicillin molecules were conjugated on the GNP surfaces, therefore considerable amount of drug was distributed on to kidney and spleen.

4. Discussion

Antibiotic discovery and development have greatly decelerated in recent years, regardless of the constantly increasing need and the deadly epidemic of multidrug-resistant bacteria. Conventional antibiotics are losing their effectiveness against infections. In this regard, researchers are trying to find out effective alternative therapeutic approaches to overcome MRSA and other resistant infections. In such an attempt, this present study demonstrates the functionalization of widely used β -lactam antibiotic on GNP surface (GNP-Amox) and evaluated the same against MRSA and other pathogenic bacterial strains. The phytochemicals present in the aqueous extract of *A. philippense* such as

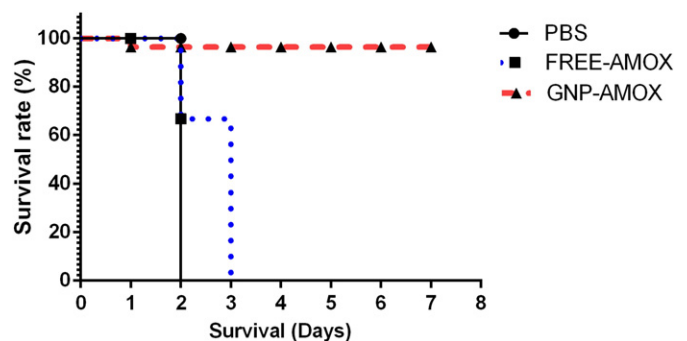


Fig. 5. Survival study of mice infected with MRSA and treated with PBS, free-Amox (50 mg/kg) and GNP-Amox (50 mg/kg). Mice were treated with each agent by duration gap of 24 h for 7 day period and the survival rate was determined by Kaplan–Meier curve.

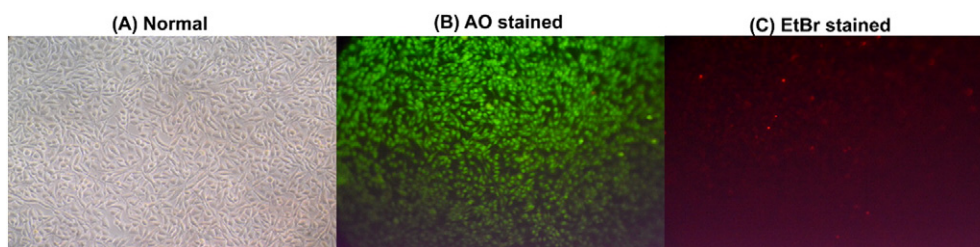


Fig. 6. Live/dead assay: Images of L929 cells (A: normal view, B: AO stained and C: EtBr stained) after treated with 250 $\mu\text{g}/\text{ml}$ of GNPs as observed under inverted fluorescent microscope (magnification $\times 200$).

terpenoids, flavonoids and alkaloids were used for synthesis of GNPs which acts both as a reducing and capping agent and thereby stabilizing the GNP solution. The physicochemical characterization of GNP confirmed the formation of stable nanoparticles. Further characterization of GNP-Amox established the formation of stable conjugates of drug and GNPs. It is reported that upon binding on to GNPs, reactive portion of the ciprofloxacin molecule was surface exposed [44]. Our study also suggests that amoxicillin functionalized on to GNP surface most likely retains its activity because bare GNP didn't exhibit antibacterial activity, whereas GNP-Amox demonstrated potent antibacterial activity against both the classes of bacteria along with MRSA. Additional studies are required to prove the retention of drug activity when it is bound to GNPs. Zone of inhibition, MIC, MBC and time kill assays confirms the revision of MRSA resistance and enhanced antibacterial activity of GNP-Amox against tested pathogenic bacterial strains. The GNP-Amox also showed promising results in clearance of MRSA infections of mice kidney and spleen resulted in significant increase in the survival rate. This study also reveals that Gram-negative bacteria were more susceptible to GNP-Amox than Gram-positive owing to the thinner Gram-negative cell walls. The reversion and enhanced effectiveness of amoxicillin conjugation on GNPs may be due to increase the concentration of antibiotics at the site of bacterium–antibiotic interaction, facilitate binding of antibiotics to the bacteria and blockade of the bacterial efflux pump in the removal of GNP-AMOX [2]. Amoxicillin inhibits the third and last stage of bacterial cell wall synthesis which results in cell lysis [45]. Upon entering in to bacterial cell, nanoparticles can alter the functions of cell membranes such as permeability and respiration as well as can disturb the functions of sulphur-containing proteins and phosphorus containing compounds such as DNA by effectively reacting with them [34,46]. It has been also reported that nanoparticles combination with antibiotics shows higher antibacterial efficacy due to its ability of protein synthesis inhibition. Furthermore, transcriptomic and proteomic studies have shown that GNPs can destabilize membrane potential by inhibiting ATPase activities to decrease the ATP level. It can also interfere with binding affinity of ribosome subunit to tRNA. Moreover, GNPs enhances chemotaxis in the early-phase reaction but are not involved in reactive oxygen species (ROS)-related mechanism which is the cause for cellular death induced by most bactericidal antibiotics and nanomaterials. The prepared GNP and GNP-Amox showed negligible cytotoxic effects on L929 mammalian cells as supported by MTT assay, live/dead cell assay and colony forming assay. ROS-independent mechanism of action of gold NPs partly explain the low toxicity of GNPs and GNP-Amox [46]. The combined use of nanoparticles with antibiotics makes it possible to reduce the toxicity of both agents towards human cells due to decreased requirement for high dosages as well as synergistically enhancing their antimicrobial activities. Metals complex action mechanisms decrease the probability of bacteria developing resistance to them [47]. Therefore it is evident that there is an urgent need of research to understand the mechanisms of action behind nanoparticle and antibiotic conjugates and its enhanced antibacterial activity. It is believed that one of the important steps in combating antibiotic-resistant bacteria is impairing the structures of proteins that are responsible for resistance. Therefore, we propose that more studies

must be focused on combinations of antibiotics with nanoparticles that strongly interact with proteins and impair their structures.

5. Conclusion

The present study reports for the first time a rapid and facile method of conjugating antibiotics on the surface of GNPs by simple electrostatic interaction. The stable functionalization of amoxicillin on GNPs revert MRSA resistance and significantly enhanced the antibacterial efficacy against tested pathogenic bacterial strains. *In vivo* studies showed decreased MRSA load in spleen and kidney as well as increased survival rate of GNP-Amox treated experimental animals. We surmise that the reversion of MRSA resistance and enhanced antimicrobial activity of GNP-Amox is due to; (A) increase in the concentration of antibiotics at the site of bacterium–antibiotic interaction (B) multivalent presentation of amoxicillin (C) blockade of the bacterial efflux pump in the removal of GNP-AMOX. However, the exact mechanism by which the GNP-Amox conjugate overcomes the antibiotic resistance displayed by MRSA strains and how it counters the effect of high levels of β -lactamase is still not clear and related research is presently underway in our laboratory. It is evident from the results obtained in the present study that GNPs play a vital role in delivering antibiotics not only as a efficient, target specific drug carrier but also in augmenting the bactericidal efficacy of the drug and hence can contribute in the development of novel therapeutic solutions against drug resistant strains of pathogenic bacteria.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.msec.2015.12.078>.

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